Hyperbola: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Recognition and definition

Formulas

$$\Delta \neq 0,\ h^2 > ab$$

$$|PS_1 - PS_2| = 2a < S_1S_2$$

$$PS = e \cdot PM,\ e > 1$$

KEY POINTS

- Rectangular if also $a + b = 0$ (coefficients of x^2, y^2)
- $2a = S_1S_2$: two rays; $2a > S_1S_2$: no locus
- Circle touching two fixed circles externally: centre on one branch, foci C_1, C_2, $2a = |r_1 - r_2|$
- $PS_1 - PS_2 = +2a$ alone gives only one branch

Watch out

Check $\Delta \neq 0$: $\Delta = 0$ with $h^2 > ab$ is a pair of intersecting lines

2. Standard hyperbola

Formulas

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \quad b^2 = a^2(e^2 - 1)$$

$$e = \sqrt{1 + \frac{b^2}{a^2}} > 1$$

KEY POINTS

- Vertices $(\pm a, 0)$, CA ends $(0, \pm b)$, foci $(\pm ae, 0)$, directrices $x = \pm\frac{a}{e}$
- LR ends $\left(\pm ae, \pm\frac{b^2}{a}\right)$, $\ell_{LR} = \frac{2b^2}{a} = 2a(e^2 - 1) = 2e\times$(focus–directrix distance)
- Vertical $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$: foci $(0, \pm ae)$, directrices $y = \pm\frac{a}{e}$
- Similar hyperbolas $\Leftrightarrow$ same e
- Order of foci, directrices, vertices on the TA: $0 < \frac{a}{e} < a < ae$

Watch out

a is under the POSITIVE term, not the bigger denominator
Directrices lie between the centre and the vertices

3. Focal distances

Formulas

$$PS_1 = |ex_1 - a|, \quad PS_2 = |ex_1 + a|$$

$$|PS_1 - PS_2| = 2a$$

KEY POINTS

- $PS_1 \cdot PS_2 = e^2x_1^2 - a^2 = CP^2 - a^2 + b^2$
- Right branch: $PS_2 - PS_1 = 2a$; left branch: $PS_1 - PS_2 = 2a$
- Area of $\triangle PS_1S_2 = \frac{1}{2} \cdot 2ae \cdot |y_1|$

4. Shifted and other forms

Formulas

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$\frac{p_2^2}{a^2} - \frac{p_1^2}{b^2} = 1$$

STEPS

1. Complete squares, put $X = x - h$, $Y = y - k$
2. Find everything in X, Y, then shift back
3. $PS = e\,PM$ form: TA $\perp$ directrix through S, $ae - \frac{a}{e}$ = focus–directrix distance
4. Alternate form: divide each bracket by $\sqrt{l^2 + m^2}$; line under a^2 = conjugate axis

Watch out

In the alternate form the transverse axis is the line in the NEGATIVE term

5. Auxiliary circle and parametric form

Formulas

$$x^2 + y^2 = a^2$$

$$P(\theta) = (a\sec\theta,\ b\tan\theta)$$

KEY POINTS

- Corresponding point on the auxiliary circle: $Q(a\cos\theta, a\sin\theta)$
- θ in quadrant 2 gives P in quadrant 3 and vice versa
- Eliminating a parameter with $\sec^2 - \tan^2 = 1$ or $\left(t + \frac{1}{t}\right)^2 - \left(t - \frac{1}{t}\right)^2 = 4$ gives a hyperbola

6. Conjugate and rectangular hyperbola

Formulas

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1, \quad \frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$$

$$x^2 - y^2 = a^2 :\ e = \sqrt{2},\ \ell_{LR} = \ell_{TA} = \ell_{CA} = 2a$$

KEY POINTS

- Conjugate: axes interchange; foci $(0, \pm be')$, $\ell_{LR} = \frac{2a^2}{b}$
- The four foci lie on $x^2 + y^2 = a^2 + b^2$ and form a square
- Rectangular $\Leftrightarrow a = b \Leftrightarrow$ asymptotes perpendicular

7. Position of a point, chord

Formulas

$$S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$$

$$\frac{x}{a}\cos\frac{\alpha - \beta}{2} - \frac{y}{b}\sin\frac{\alpha + \beta}{2} = \cos\frac{\alpha + \beta}{2}$$

KEY POINTS

- $S_1 > 0$ inside (focus side), $= 0$ on, < 0 outside (centre side)
- Chord through $(d, 0)$: $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{a-d}{a+d}$; focal chord: $\frac{1-e}{1+e}$

8. Line and hyperbola, director circle

Formulas

$$c^2 = a^2m^2 - b^2$$

$$y = mx \pm \sqrt{a^2m^2 - b^2}$$

$$x^2 + y^2 = a^2 - b^2$$

KEY POINTS

- $c^2 > a^2m^2 - b^2$: two points; $<$: none; tangents need $|m| > \frac{b}{a}$

- Line parallel to an asymptote meets the curve once (not a tangent)
- Signed form $\frac{x^2}{A} - \frac{y^2}{B} = 1$: $c^2 = Am^2 - B$ (conjugate: $c^2 = b^2 - a^2m^2$)
- From (h, k): $(h^2 - a^2)m^2 - 2hkm + k^2 + b^2 = 0$
- Director circle real only for $a > b$; point circle for $a = b$

9. Tangent

Formulas

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

$$\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$$

$$\text{contact of } y = mx + c: \left(-\frac{a^2m}{c}, -\frac{b^2}{c}\right)$$

KEY POINTS

- Tangents at α, β meet at $\left(\frac{a\cos\frac{\alpha-\beta}{2}}{\cos\frac{\alpha+\beta}{2}}, \frac{b\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha+\beta}{2}}\right)$
- Slope at an LR end $= \pm e$; $CT \cdot CN = a^2$
- Foot of $\perp$ from centre on tangent: $(x^2 + y^2)^2 = a^2x^2 - b^2y^2$
- Line to curve least distance: parallel tangent nearer to the line

10. Normal

Formulas

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$$

$$ax\cos\theta + by\cot\theta = a^2 + b^2$$

$$c^2 = \frac{m^2(a^2 + b^2)^2}{a^2 - b^2m^2}$$

KEY POINTS

- Normal meets TA at $(e^2x_1, 0)$, CA at $\left(0, \frac{a^2+b^2}{b^2}y_1\right)$
- At most four normals from a point; shortest distance from a point is along a normal

11. Chords, pole and polar

Formulas

$$T = S_1, \quad T = 0, \quad SS_1 = T^2$$

$$\text{pole of } lx + my + n = 0: \left(-\frac{a^2l}{n}, \frac{b^2m}{n}\right)$$

$$\text{diameter: } y = \frac{b^2}{a^2m}x$$

KEY POINTS

- $T = S_1$: chord with given mid-point; locus of mid-points = impose condition on it
- $T = 0$: tangent / chord of contact / polar
- Point on a fixed line $\Rightarrow$ chord of contact through the pole of that line

12. Highlights

KEY POINTS

- $p_1p_2 = b^2$ (perpendiculars from foci on a tangent)
- Foot of $\perp$ from a focus on a tangent lies on $x^2 + y^2 = a^2$
- Tangent between point of contact and directrix subtends $90°$ at the focus
- Circle on a focal distance as diameter touches the auxiliary circle

- Tangent bisects the angle between the focal radii (reflection: ray towards one focus goes to the other)
- Confocal ellipse and hyperbola cut at right angles

13. Asymptotes

Formulas

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0, \quad H + C = 2A$$

$$\tan\theta = \frac{2ab}{a^2 - b^2}, \quad e = \sec\frac{\theta}{2}$$

KEY POINTS

- General hyperbola: $A = H + \lambda$ with $\Delta = 0$; conjugate $C = 2A - H$
- Asymptotes $L_1L_2 = 0$ given: hyperbola $L_1L_2 = k$
- Axes bisect the angles between asymptotes; focus is at distance b from each asymptote
- Tangent at P meets asymptotes at Q, R: P = mid-point, area $CQR = ab$
- Tangents from a point: same branch / one to each branch / one (on an asymptote) / none (centre)

14. Rectangular hyperbola $xy = c^2$

Formulas

$$P(t) = \left(ct, \frac{c}{t}\right)$$

$$x + t^2y = 2ct, \quad t^3x - ty = c(t^4 - 1)$$

$$x + t_1t_2y = c(t_1 + t_2), \quad \frac{x}{h} + \frac{y}{k} = 2$$

KEY POINTS

- Vertices $\pm(c, c)$, foci $\pm(\sqrt{2}c, \sqrt{2}c)$, directrices $x + y = \pm\sqrt{2}c$, $\ell_{LR} = 2\sqrt{2}c$
- Normal at t meets again at t_1: $t^3t_1 = -1$
- $y = mx + k$ touches iff $k^2 + 4mc^2 = 0$
- Circle meets it at $t_1..t_4$: $t_1t_2t_3t_4 = 1$, $\sum x_i = -2g$, $\sum y_i = -2f$

15. Common mistakes checklist

STEPS

1. $b^2 = a^2(e^2 - 1)$, not $a^2(1 - e^2)$
2. Tangent condition $c^2 = a^2m^2 - b^2$ (minus sign); conjugate: $c^2 = b^2 - a^2m^2$
3. Vertical hyperbola: foci on the y-axis even if $a < b$
4. $|PS_1 - PS_2| = 2a$ uses the absolute value for both branches
5. Director circle does not exist when $b > a$
6. Asymptote of $xy = c^2$ are the axes; its $a = \sqrt{2}c$, not c